Math 4261: Mathematical Statistics I

Lecture 17: Parameter estimation
under model misspecification
(References: C. Geyer[1], Z. Fan[2])

Fall 2026

[1] https://www.stat.umn.edu/geyer/5601/notes/sand.pdf

[2] https://web.stanford.edu/class/archive/stats/stats200/stats200.1172/Lecture16.pdf

Outline

Recap

Model misspecification

Kullback-Leibler divergence

MLE in misspecified models

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Recap: Bayesian credible sets

A Bayesian credible interval $I_{\text{Bayes}}(\boldsymbol{X})$ with 95% coverage is an interval which guarantees

$$\mathbb{P}[\Theta \in I_{\text{Bayes}}(\boldsymbol{X}) \mid \boldsymbol{X} = \boldsymbol{x}] = 0.95.$$

- The randomness is over Θ, given $\boldsymbol{X} = \boldsymbol{x}$.
- Contrast to frequentist confidence interval which guarantees for a *fixed* θ that $\mathbb{P}_\theta[\theta \in I(\boldsymbol{X})] = 95\%$.

  This is the probability that $I(\boldsymbol{X}) \ni \theta$, over the randomness of $\boldsymbol{X} \sim f(\boldsymbol{x} \mid \theta)$.

OPTIONAL

- The $I_{\text{Bayes}}(\boldsymbol{X})$ does not guarantee coverage in the frequentist sense for any fixed θ, but only on average:

$$95\% = \int \mathbb{P}_\theta[\theta \in I_{\text{Bayes}}(\boldsymbol{X})]\, f_\Theta(\theta)\, \mathrm{d}\theta$$

Recap: Theoretical aspects of Bayesian inference

Best estimators

Let X and Y be any two random variables.

- We want to use X (observed) to predict Y (unobserved).
- Call our estimator $\delta(X)$.
- A common measure of the quality of our estimate is Mean-square error (MSE) = $\mathbb{E}\left[(\delta(X) - Y)^2\right]$.
- The best estimator, the one that minimizes MSE, is the conditional expectation

$$\delta(X) = \mathbb{E}[Y \mid X]$$

Linear regression is in some sense, restricting to estimators $\delta(\boldsymbol{X})$ that are linear functions of $\boldsymbol{X}$.

Recap: Theoretical aspects of Bayesian inference

Translation to Bayesian setting
Generalizing the above:

- Θ is our unknown random parameter
- Data $\boldsymbol{X} \sim f_{\boldsymbol{X}|\Theta}$.
- The best estimator, the one minimizing the MSE

$$\mathrm{MSE}(\delta) = \mathbb{E}\left[(\Theta - \delta(\boldsymbol{X}))^2\right]$$

over all possible estimators δ, is the **posterior mean**

$$\delta_{\mathrm{Bayes}}(\boldsymbol{X}) = \mathbb{E}[\Theta \mid \boldsymbol{X}].$$

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Model misspecification

Up till this point, we typically start with the assumption

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta).$$

Here $f(x \mid \theta)$ plays two different roles:

- It describes the model we choose to *fit* to the data
- It describes the *true* distribution of the data

In reality these can be **different**, for various reasons:

- The true distribution of the data is unknown, and any model we fit is only an approximation to the truth
- The data is well-described by a complex model, but we choose to fit a simpler model that has more interpretable parameters
- We fit a model that is only intended for black-box prediction, rather than accurately describing the data-generating process

Q: How may we interpret the fitted parameter $\widehat{\theta}$ and model $f(x \mid \widehat{\theta})$ if $f(x \mid \theta)$ does **not** truly describe the distribution of the data?

Example: Interpreting linear regression

Suppose $(X_1, Y_1), \ldots, (X_n, Y_n) \in \mathbb{R}^2$ are covariate-response[3] pairs, which are i.i.d. with some unknown distribution $g(x, y)$.
E.g. X = height of parent, Y = height of child.

Insights from the best estimator calculation tells us that ideally we use $\mathbb{E}[Y \mid X]$ to predict Y. In **linear regression** we just assume

$$\mathbb{E}[Y \mid X] \approx \beta_0 + \beta_1 X, \quad \beta_0, \beta_1 \in \mathbb{R}$$

One interpretation is that we are positing a model

$$Y = \beta_0 + \beta_1 X + \text{Gaussian noise}$$

We do **not** actually believe that the true distribution g is of this form. This is a modeling assumption. If g is not actually of this form, then this is an example of model misspecification.

[3] Recall the best linear prediction setting from Lecture 4.

Example: Interpreting linear regression

We estimate the parameters $\beta = (\beta_0, \beta_1)$ using least squares:

$$(\widehat{\beta}_0, \widehat{\beta}_1) = \arg\min_{\beta_0, \beta_1} \frac{1}{n} \sum_{i=1}^{n} (\beta_0 + \beta_1 X_i - Y_i)^2.$$

We may interpret $(\widehat{\beta}_0, \widehat{\beta}_1)$ as representing the "best linear predictor" for Y based on X. For large n, we expect $(\widehat{\beta}_0, \widehat{\beta}_1)$ to be close to the minimizers of the (population) mean squared error

$$\arg\min_{\beta_0, \beta_1} \mathbb{E}_{(X,Y)\sim g}\Big[(\beta_0 + \beta_1 X - Y)^2\Big].$$

Example: Interpreting MoM under misspecification

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} g(x)$, for an unknown distribution $g(x)$.

We fit a model $f(x \mid \theta)$ with parameters $\theta \in \Omega \subseteq \mathbb{R}^k$ using the **method of moments**. By definition, the estimated parameters $\widehat{\theta}$ are those for which

$$\mathbb{E}_{\widehat{\theta}}[X] = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad \ldots, \quad \mathbb{E}_{\widehat{\theta}}[X^k] = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

This may be understood as estimating the distribution $f(x \mid \theta)$ within our model that matches $g(x)$ in its first k moments.

For large n, we expect $\widehat{\theta}$ to be close to the value of θ such that

$$\mathbb{E}_{\theta}[X] = \mathbb{E}_{X\sim g}[X], \quad \ldots, \quad \mathbb{E}_{\theta}[X^k] = \mathbb{E}_{X\sim g}[X^k].$$

Model misspecification

Usual MLE

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta_0)$. Fit a model $\{f(x \mid \theta) : \theta \in \Omega\}$. MLE $\widehat{\theta}$ is maximizer of log-likelihood.

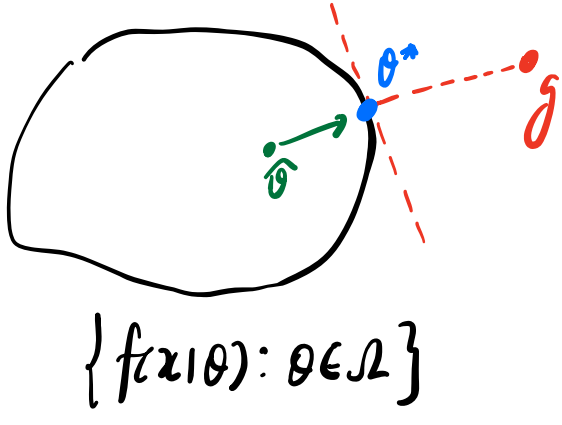


Eg. $\{\text{Poisson}(\lambda) : \lambda > 0\}$

Theoretical result:
$\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \frac{1}{I(\theta_0)})$.

General setting

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} g(x)$. Fit a model $\{f(x \mid \theta) : \theta \in \Omega\}$. MLE $\widehat{\theta}$ is maximizer of log-likelihood.



The appropriate 'distance' is the **Kullback-Leibler divergence**.

Main theorem on misspecified models

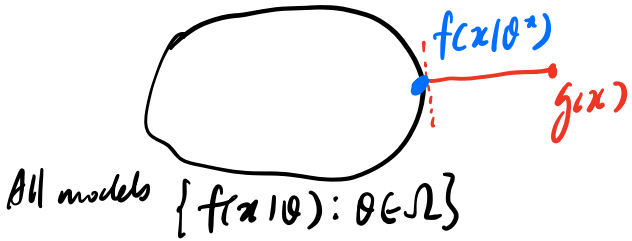
Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} g(x)$, for an unknown distribution $g(x)$. We fit a model $f(x \mid \theta)$ over parameters $\theta \in \Omega$ using **maximum likelihood**. What is the interpretation of $\widehat{\theta}$?

Theorem

Let $D_{\mathrm{KL}}(g(x) \,\|\, f(x \mid \theta))$ be the Kullback–Leibler (KL) divergence of $g(x)$ relative to $f(x \mid \theta)$, and suppose that

$$\theta \mapsto D_{\mathrm{KL}}(g(x) \,\|\, f(x \mid \theta))$$

has a unique minimizer $\theta^* \in \Omega$. Then, under regularity conditions for $f(x \mid \theta)$ and g, the MLE $\widehat{\theta}$ converges to θ^* in probability as $n \to \infty$.



For large n, the MLE estimates the "KL projection" of the true distribution $g(x)$ onto our model $\{f(x \mid \theta) : \theta \in \Omega\}$.

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Motivation

Thus far we have been measuring the error of an estimator $\widehat{\theta}$ by its *distance* to the true parameter θ, via the bias, variance, and MSE.

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} g$ for a PDF/PMF g that is **not** in the model $\{f(x \mid \theta) : \theta \in \Omega\}$ that we are considering, then there is no true parameter value θ associated to g.

To understand how well our model approximates the true distribution, we will instead need a measure of "distance" between two general PMFs/PDFs.

Kullback–Leibler (KL) divergence

For two discrete distributions with PMFs f and g on a sample space $\mathcal{X}$, the **Kullback–Leibler (KL) divergence** (of g relative to f) is

$$D_{\mathrm{KL}}(g \,\|\, f) = \sum_{x \in \mathcal{X}} g(x) \log \frac{g(x)}{f(x)}.$$

For two continuous distributions with PDFs f and g on the real line, the KL divergence (of g relative to f) is similarly

$$D_{\mathrm{KL}}(g \,\|\, f) = \int_{-\infty}^{\infty} g(x) \log \frac{g(x)}{f(x)} \, dx.$$

In both cases, this may be written as an expectation

$$\boxed{D_{\mathrm{KL}}(g \,\|\, f) = \mathbb{E}_g\left[\log \frac{g(X)}{f(X)}\right] = \mathbb{E}_g[\log g(X)] - \mathbb{E}_g[\log f(X)],}$$

where $\mathbb{E}_g$ indicates expectation with respect to $X \sim g(x)$.

Example: KL between Gaussians

Let $f = \mathcal{N}(\mu_0, \sigma^2)$ and $g = \mathcal{N}(\mu_1, \sigma^2)$ with common variance σ^2.
Compute $D_{\mathrm{KL}}(g \,\|\, f)$:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x-\mu_0)^2}{2\sigma^2}\right), \quad g(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right)$$

Hence $$\log \frac{g(x)}{f(x)} = -\frac{(x-\mu_1)^2}{2\sigma^2} + \frac{(x-\mu_0)^2}{2\sigma^2}$$

$$= -\frac{1}{2\sigma^2}x^2 + \frac{1}{\sigma^2}x\mu_1 - \frac{1}{2\sigma^2}\mu_1^2$$

$$+ \frac{1}{2\sigma^2}x^2 - \frac{1}{\sigma^2}x\mu_0 + \frac{1}{2\sigma^2}\mu_0^2$$

$$= \frac{\mu_1 - \mu_0}{\sigma^2}x - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2}$$

Example: KL between Gaussians

$$D_{KL}(g\|f) = \mathbb{E}_g \log \frac{g(x)}{f(x)}$$

$$= \mathbb{E}_g\left[\frac{\mu_1-\mu_0}{\sigma^2}X - \frac{\mu_1^2-\mu_0^2}{2\sigma^2}\right]$$

$$= \frac{\mu_1-\mu_0}{\sigma^2}\mu_1 - \frac{\mu_1^2-\mu_0^2}{2\sigma^2}$$

$$= \frac{2\mu_1^2 - 2\mu_0\mu_1 - \mu_1^2 + \mu_0^2}{2\sigma^2} = \frac{(\mu_1-\mu_0)^2}{2\sigma^2}$$

Observe For $X\sim N(\mu,\sigma^2)$, known σ^2,

$$I(\mu) = -\mathbb{E}\left[\frac{\partial^2}{\partial\mu^2}\log\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}\right] = \dots = \frac{1}{\sigma^2}$$

so $$\boxed{D_{KL}(g\|f) = \frac{I(\mu_1)}{2}(\mu_1-\mu_0)^2}$$

Example: KL between Bernoulli's

PMF $g(x) = \mathbb{P}[X = x]$

Let f = Bernoulli(p), g = Bernoulli(q), where $p, q \in (0,1)$. Compute $D_{\mathrm{KL}}(g \,\|\, f)$:

$$\log \frac{g(x)}{f(x)} = \begin{cases} \log \frac{q}{p} & \text{if } x = 1 \\ \log \frac{1-q}{1-p} & \text{if } x = 0 \end{cases}$$

$$D_{KL}(g \| f) = \mathbb{E}_g\left[\log \frac{g(X)}{f(X)}\right]$$

$$= q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}$$

Note: This expression is not symmetric in (p,q).

So $D_{KL}(g \| f) \neq D_{KL}(f \| g)$.

Example: KL between Bernoulli's

For $f = \text{Bernoulli}(p)$ and $g = \text{Bernoulli}(q)$, we have

$$\boxed{D_{\mathrm{KL}}(g \,\|\, f) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}.}$$

For p close to q, this may be approximated by Taylor expansion.
Recall $\log(1+x) \approx x - \frac{x^2}{2}$ for $|x| \approx 0$.

Let $\Delta = q - p$, assume $|\Delta| \approx 0$

$$\log \frac{q}{p} = \log\left(1 + \left(\frac{q}{p} - 1\right)\right) = \log\left(1 + \frac{\Delta}{p}\right)$$

$$\approx \frac{\Delta}{p} - \frac{\Delta^2}{2p^2}$$

Example: KL between Bernoulli's

$\Delta = q - p$, $|\Delta| \approx 0$

$q = p + \Delta$:

$$q \log \frac{q}{p} \approx (p+\Delta)\left(\frac{\Delta}{p} - \frac{\Delta^2}{2p^2}\right)$$

$$= \Delta - \frac{\Delta^2}{2p} + \frac{\Delta^2}{p} - \frac{\Delta^3}{2p^2}$$

smaller order, drop

$$= \Delta + \frac{\Delta^2}{2p}.$$

Similarly, $(1-p)\log\frac{1-q}{1-p} \approx -\Delta + \frac{\Delta^2}{2(1-p)}$

Thus $D_{KL}(q\|p) \approx \frac{\Delta^2}{2p} + \frac{\Delta^2}{2(1-p)} = \frac{1}{2p(1-p)}(q-p)^2 \approx \frac{1}{2q(1-q)}(q-p)^2$

Observe $X \sim \text{Bernoulli}(q) = q \Rightarrow I(q) = \frac{1}{q(1-q)}$

$= \frac{I(q)}{2}(q-p)^2$

Example: KL between Binomials

Let $f = \text{Binomial}(n, p)$ and $g = \text{Binomial}(n, q)$, where $p, q \in (0, 1)$. Then

$$\log \frac{g(x)}{f(x)} = \log \left(\frac{\binom{n}{x} q^x (1-q)^{n-x}}{\binom{n}{x} p^x (1-p)^{n-x}} \right) = x \log \frac{q}{p} + (n - x) \log \frac{1-q}{1-p}.$$

Applying $\mathbb{E}_g[X] = nq$,

$$\begin{aligned} D_{\mathrm{KL}}(g \,\|\, f) &= \mathbb{E}_g[X] \log \frac{q}{p} + (n - \mathbb{E}_g[X]) \log \frac{1-q}{1-p} \\ &= n \left(q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p} \right) \\ &= \boxed{n \times D_{\mathrm{KL}}(\text{Bernoulli}(q) \,\|\, \text{Bernoulli}(p)).} \end{aligned}$$

For p close to q, this is

$$D_{\mathrm{KL}}(g \,\|\, f) \approx n \frac{(p-q)^2}{2q(1-q)}.$$

Properties of KL divergence

$D_{\mathrm{KL}}(g \,\|\, f)$ is an information-theoretic measure of discrepancy between g and f. In general, it is not-symmetric:

$$D_{\mathrm{KL}}(f \,\|\, g) \neq D_{\mathrm{KL}}(g \,\|\, f).$$

It satisfies the following **basic properties**:

- If $f = g$, then $D_{\mathrm{KL}}(g \,\|\, f) = 0$ because $\log \frac{g(x)}{f(x)} = 0$ for all x.
- $D_{\mathrm{KL}}(g \,\|\, f) \geq 0$ for any f and g. This follows from applying Jensen's inequality to the convex function $-\log x$.

$$D_{KL}(g\|f) = \mathbb{E}_g\left[\log \frac{g(x)}{f(x)}\right] = \mathbb{E}_g\left[-\log \frac{f(x)}{g(x)}\right]$$

$$\geq -\log \mathbb{E}_g\left[\frac{f(x)}{g(x)}\right] = -\log \underbrace{\int_{\mathbb{R}} \frac{f(x)}{g(x)}\, g(x)\, dx}_{=1}$$

$$= 0$$

KL divergence within a parametric model

Consider a parametric model $f(x \mid \theta)$ and two parameters $\theta_0, \theta_1 \in \Omega \subseteq \mathbb{R}$. Then

$$D_{\mathrm{KL}}(f(x \mid \theta_0) \,\|\, f(x \mid \theta_1)) = \mathbb{E}_{\theta_0}[\log f(X \mid \theta_0)] - \mathbb{E}_{\theta_0}[\log f(X \mid \theta_1)].$$

Fixing θ_0, let

$$L(\theta) = \mathbb{E}_{\theta_0}[\log f(X \mid \theta)]$$

maximized at $\theta = \theta_0$

be the population log-likelihood function when the true parameter is θ_0. If θ_0 and θ_1 are close to each other, then we may apply the Taylor expansion

$$L(\theta_1) \approx L(\theta_0) + (\theta_1 - \theta_0)L'(\theta_0) + \frac{1}{2}(\theta_1 - \theta_0)^2 L''(\theta_0).$$

Recall that

- $L'(\theta_0) = 0$ because θ_0 maximizes $L(\theta)$
- $L''(\theta_0) = -I(\theta_0)$, the Fisher information at θ_0

KL divergence within a parametric model

Thus, for parameters θ_0 and θ_1 close to each other,

$$\boxed{D_{\mathrm{KL}}\big(f(x \mid \theta_0) \,\|\, f(x \mid \theta_1)\big) \approx \frac{I(\theta_0)}{2}(\theta_1 - \theta_0)^2.}$$

- The KL divergence behaves (sort of) like the squared distance.
- This gives another interpretation of the Fisher information: It relates the KL divergence — an information-theoretic measure of discrepancy between $f(x \mid \theta_0)$ and $f(x \mid \theta_1)$ — to the squared difference of the model parameters.

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Interpretation of the MLE

(Restated) Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} g(x)$, for an unknown distribution $g(x)$.

We fit a model $f(x \mid \theta)$ over parameters $\theta \in \Omega$ using **maximum likelihood**. What is the interpretation of $\widehat{\theta}$?

Theorem

Let $D_{\text{KL}}(g(x) \,\|\, f(x \mid \theta))$ be the Kullback–Leibler (KL) divergence of $g(x)$ relative to $f(x \mid \theta)$, and suppose that

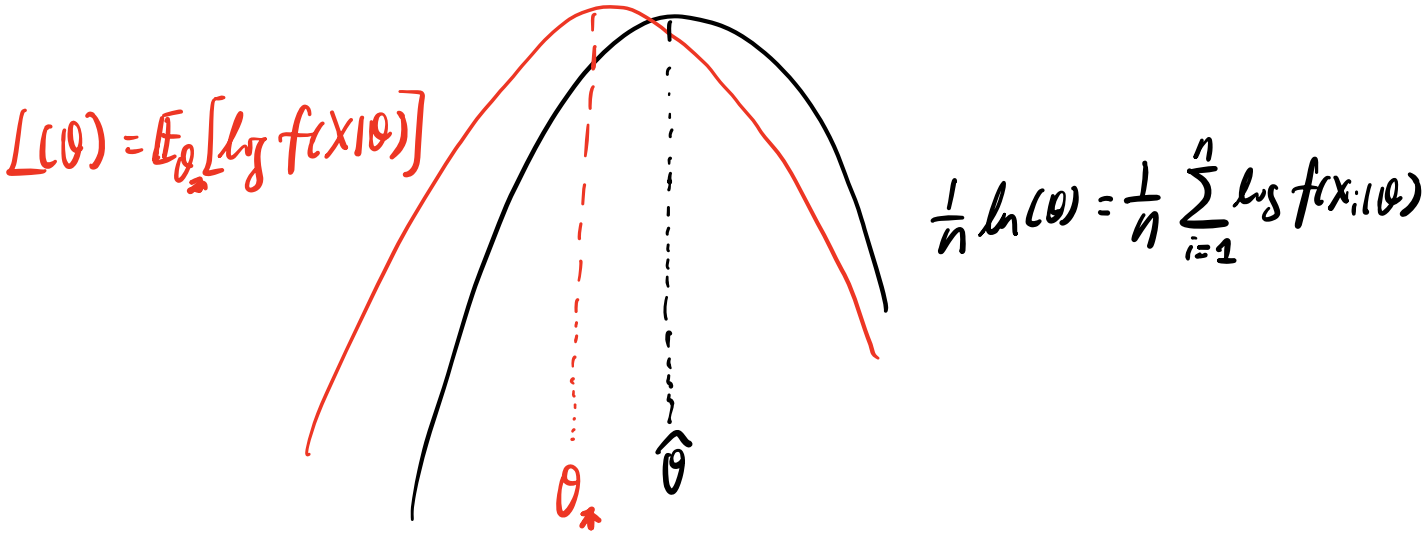
$$\theta \mapsto D_{\text{KL}}(g(x) \,\|\, f(x \mid \theta))$$

has a unique minimizer $\theta^* \in \Omega$. Then, under regularity conditions[a] for $f(x \mid \theta)$ and g, the MLE $\widehat{\theta}$ converges to θ^* in probability as $n \to \infty$.

[a] See e.g. Halbert White (1982) "Maximum likelihood estimation of misspecified models"

Recap: Expected log-likelihood function

Usual MLE setting



If the true parameter is θ_*, then by the LLN, the average log-likelihood approaches the expected log-likelihood

$$\frac{1}{n}\ell_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i \mid \theta) \to L(\theta) = \mathbb{E}_{\theta_*}[\log f(X \mid \theta)]$$

in probability. The expected log-likelihood $L(\theta)$ is maximized at the true parameter θ_*.

Proof sketch

By definition, MLE $\hat{\theta}$ maximizes

$$\frac{1}{n}\ell_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i|\theta)$$

By LLN, for each fixed θ,

$$\frac{1}{n}\ell(\theta) \xrightarrow{\text{prob}} \underbrace{\mathbb{E}_g[\log f(X|\theta)]}_{\text{call this } L(\theta)}, \quad \text{here } X \sim g$$

Previously, assumed $g = f(x|\theta_*)$ for some true θ_*.

Then $L(\theta)$ maximized at θ_*. (yields consistency)

Proof sketch

If $g(x)$ is not in the model $\{f(x|\theta) : \theta \in \Omega\}$,

write

$$L(\theta) = \mathbb{E}_g[\log f(x|\theta)]$$

$$= \mathbb{E}_g \log\left(g(x) \cdot \frac{f(x|\theta)}{g(x)}\right)$$

$$= \underbrace{\mathbb{E}_g[\log g(x)]}_{\text{does not involve } \theta} - \underbrace{\mathbb{E}_g \log \frac{g(x)}{f(x|\theta)}}_{D_{KL}(g(x) \,\|\, f(x|\theta))}$$

Thus

maximizer $L(\theta)$ $\longleftrightarrow$ minimizer D_{KL} (a unique $\theta^* \in \Omega$)

and $\hat{\theta} \to \theta^*$ in prob.

Example: MLE in a misspecified Poisson model

Suppose we fit the model Poisson(λ) to nonnegative integer-valued observations $X_1, \ldots, X_n$ whose true distribution is $g(x)$. Assume $0 < \mathbb{E}_g[X] < \infty$ and $\mathrm{Var}_g[X] < \infty$.

Our model: $f(x|\lambda) = \dfrac{e^{-\lambda}\lambda^x}{x!}$ PMF of Poisson(λ)

$$D_{KL}(g(x) \| f(x|\lambda)) = \mathbb{E}_g\left[\log \frac{g(X)}{f(X|\lambda)}\right]$$

$$= \mathbb{E}_g[\log g(X)] - \mathbb{E}_g[-\lambda + X\log\lambda - \log(X!)]$$

$$= \mathbb{E}_g[\log g(X)] + \lambda - \mathbb{E}_g[X]\cdot\log\lambda + \mathbb{E}_g[\log(X!)]$$

Example: MLE in a misspecified Poisson model

$$\frac{d}{d\lambda} D_{KL}(g(x) \| f(x|\lambda)) = 1 - \frac{\mathbb{E}_g[X]}{\lambda} \overset{\text{set}}{=} 0$$

So minimizer of $D_{KL}(g(x) \| f(x|\lambda))$ is $\lambda^* = \mathbb{E}_g[X]$

Conclude that MLE satisfies

$$\hat{\lambda} \to \lambda^* = \mathbb{E}_g[X] \quad \text{in prob.}$$

This agrees with $\hat{\lambda} = \bar{X}$ and $\bar{X} \xrightarrow{\text{in prob}} \mathbb{E}_g X$ by LLN.

Example: Asymptotic variance of the MLE

Continuing this example, since $\widehat{\lambda} = \bar{X}$, by the CLT we have

$$\sqrt{n}(\widehat{\lambda} - \lambda^*) \longrightarrow \mathcal{N}(0, \mathrm{Var}_g[X])$$

in distribution as $n \to \infty$. So

$$\mathrm{Var}_g[\widehat{\lambda}] = \frac{\mathrm{Var}_g[X]}{n}.$$

If we estimate this variance based on the Fisher information in our assumed Poisson(λ) model, we would in general obtain an incorrect estimate:

$$I(\lambda) = \mathrm{Var}_\lambda\left[\frac{\partial}{\partial\lambda}\log\frac{e^{-\lambda}\lambda^X}{X!}\right] = \mathrm{Var}_\lambda\left[\frac{X}{\lambda} - 1\right] = \frac{1}{\lambda}.$$

Thus, since $\widehat{\lambda} \approx \lambda^* = \mathbb{E}_g[X]$ for large n, the usual plug-in estimate for $\mathrm{Var}[\widehat{\lambda}]$ would be

$$\mathrm{Var}\,\widehat{\lambda} \approx \frac{1}{nI(\widehat{\lambda})} = \frac{\widehat{\lambda}}{n} \approx \frac{\lambda^*}{n} = \frac{\mathbb{E}_g[X]}{n} \underset{\text{in general}}{\neq} \frac{\mathrm{Var}_g[X]}{n}.$$

in general

Asymptotic variance for misspecified MLE

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} g(x)$. Fit a model $f(x \mid \theta)$ over parameters $\theta \in \Omega$ using maximum likelihood.
Define the score function

$$z(x, \theta) = \frac{\partial}{\partial \theta} \log f(x \mid \theta)$$

and set $z'(X, \theta) = \frac{\partial}{\partial \theta} z(X, \theta)$.

Theorem "Sandwich theorem"

Under suitable regularity conditions, let $\widehat{\theta}$ and θ^* denote the MLE and its KL-projection, where θ^* is an interior point of $\Omega \subseteq \mathbb{R}$. Then

$$\sqrt{n}\,(\widehat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\mathrm{Var}_g\left[z(X, \theta^*)\right]}{\left(\mathbb{E}_g\left[z'(X, \theta^*)\right]\right)^2}\right).$$

Here Var_g and $\mathbb{E}_g$ are with respect to $X \sim g$.

Asymptotic variance for misspecified MLE

$$\sqrt{n}\,(\widehat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\mathrm{Var}_g\left[z(X,\theta^*)\right]}{\left(\mathbb{E}_g\left[z'(X,\theta^*)\right]\right)^2}\right)$$

- In the usual MLE setting we would have

$$\sqrt{n}\,(\widehat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{1}{I(\theta^*)}\right)$$

- When $g = f(x \mid \theta^*)$, i.e. when the model is correct, then both statements coincide. (Check this).
- Proof analogous to the usual MLE setting.
- To make this usable, we still have to estimate $\mathrm{Var}_g\left[z(X,\theta^*)\right]$ and $\mathbb{E}_g\left[z'(X,\theta^*)\right]$ from the data.
- (Not important:) This is called the sandwich estimate, because in a multivariate incarnation, the variance will show up as

$$\left(\mathbb{E}_g\left[z'(X,\theta^*)\right]\right)^{-1} \mathrm{Var}_g\left[z(X,\theta^*)\right] \left(\mathbb{E}_g\left[z'(X,\theta^*)\right]\right)^{-1}$$

Summarizing remarks

- It is often meaningful to fit a parametric model $f(x \mid \theta)$ that is only an approximation to the true distribution of data.
- If maximum likelihood is used to fit the model, then $\widehat{\theta}$ may be understood as estimating a parameter θ^* representing the KL projection of the data distribution onto the assumed model.
- The asymptotic Normal approximation to $\widehat{\theta}$ has variance given by the sandwich formula:

$$\frac{\mathrm{Var}_g\left[z(X, \theta^*)\right]}{n\left(\mathbb{E}_g\left[z'(X, \theta^*)\right]\right)^2}$$

In general this is **not** the same as $\frac{1}{nI(\theta^*)}$.